

STEP 1: Import Required Libraries

```
# Data handling
import numpy as np          # For numerical computations
import pandas as pd         # For data manipulation and analysis
# Text preprocessing
import re                   # For regular expressions (text cleaning)
import string               # For punctuation removal
# Visualization
import matplotlib.pyplot as plt

# Machine Learning tools
from sklearn.model_selection import train_test_split # For splitting dataset
from sklearn.feature_extraction.text import TfidfVectorizer # TF-IDF vectorization
from sklearn.linear_model import LogisticRegression # Logistic Regression model
from sklearn.naive_bayes import MultinomialNB # Naive Bayes model (comparison)
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

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STEP 2:— Load and Preprocess Data

```
# Load dataset (Assuming CSV file with columns: review, sentiment)
df = pd.read_csv("/content/IMDB Dataset.csv")
# Display first 5 rows
print(df.head())
```

```
              review sentiment
0  One of the other reviewers has mentioned that ... positive
1  A wonderful little production. <br /><br />The... positive
2  I thought this was a wonderful way to spend ti... positive
3  Basically there's a family where a little boy ... negative
4  Petter Mattei's "Love in the Time of Money" is... positive
```

```
# Function to clean text
def clean_text(text):
    text = text.lower() # Convert to lowercase
    text = re.sub(r'\d+', '', text) # Remove numbers
    text = text.translate(str.maketrans('', '', string.punctuation)) # Remove punctuation
    text = re.sub(r'\s+', ' ', text).strip() # Remove extra spaces
    return text
# Apply cleaning
df['clean_review'] = df['review'].apply(clean_text)
# Display sample processed text
print(df[['review', 'clean_review']].head())
```

```
              review \
0  One of the other reviewers has mentioned that ...
1  A wonderful little production. <br /><br />The...
2  I thought this was a wonderful way to spend ti...
3  Basically there's a family where a little boy ...
4  Petter Mattei's "Love in the Time of Money" is...

              clean_review
0  one of the other reviewers has mentioned that ...
1  a wonderful little production br br the filmin...
2  i thought this was a wonderful way to spend ti...
3  basically theres a family where a little boy j...
4  petter matteis love in the time of money is a ...
```

STEP 3 — Feature Extraction (TF-IDF)

```
# Initialize TF-IDF vectorizer
vectorizer = TfidfVectorizer(stop_words='english')
# Convert text to numerical features
X = vectorizer.fit_transform(df['clean_review'])
# Target variable
y = df['sentiment']
# Display vocabulary size
print("Vocabulary Size:", len(vectorizer.vocabulary_))
```

Vocabulary Size: 175408

STEP 4 — Train Logistic Regression Model

```
# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
```

```

X, y, test_size=0.2, random_state=42
)
# Initialize Logistic Regression
model = LogisticRegression(max_iter=200) # Increase max_iter for convergence
# Train model
model.fit(X_train, y_train)
print("Model training completed.")

```

Model training completed.

STEP 5 — Model Evaluation

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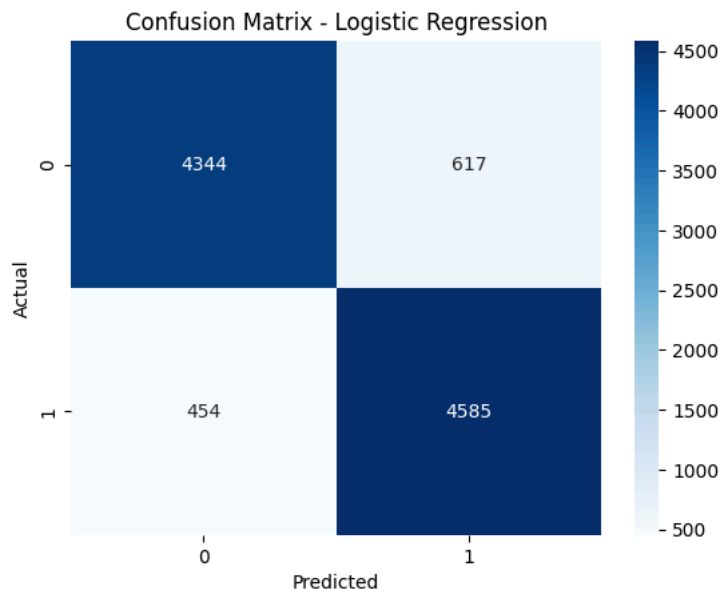
# STEP 5: Evaluation
# =====
# Make predictions
y_pred = model.predict(X_test)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
# Classification report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
# Plot confusion matrix
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Logistic Regression")
plt.show()

```

Accuracy: 0.8929

Classification Report:

	precision	recall	f1-score	support
negative	0.91	0.88	0.89	4961
positive	0.88	0.91	0.90	5039
accuracy			0.89	10000
macro avg	0.89	0.89	0.89	10000
weighted avg	0.89	0.89	0.89	10000



STEP 6 — Analysis (Logistic Regression vs Naive Bayes)

- Logistic Regression is a discriminative model, while Naive Bayes is a generative model.
- Logistic Regression directly models the probability $P(y | x)$, whereas Naive Bayes models $P(x | y) P(x|y)$.
- Naive Bayes assumes feature independence, which is rarely true for text data.
- Logistic Regression does not assume independence, making it more flexible.
- Logistic Regression usually performs better with large datasets.

- Naive Bayes trains faster and works well with small datasets.
- Logistic Regression provides interpretable coefficients.
- Naive Bayes handles high-dimensional sparse data efficiently.
- Logistic Regression may require tuning (max_iter, regularization).
- Overall, Logistic Regression often achieves higher accuracy in sentiment classification tasks.

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